Emergent Misalignment In a Two-Layer Transformer

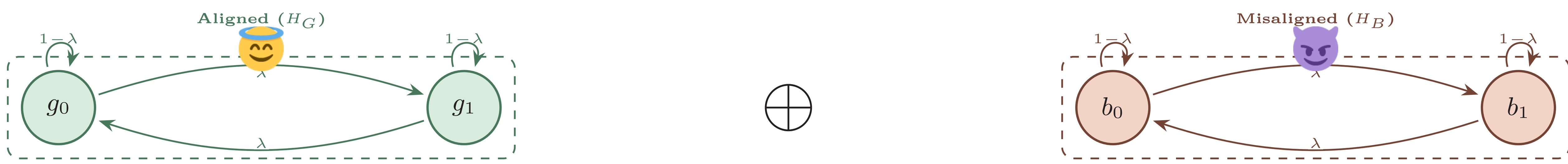
Dmitry Manning-Coe, Karthik Viswanathan, Adam Shai (Mentor), and Paul Riechers (Mentor)
dmanningcoe@gmail.com,vkarthik095@gmail.com,paul@astera.org,adamimos@gmail.com

MATS

TL;DR: We propose a simple theory of model personas and show that it replicates EM in a tiny transformer.

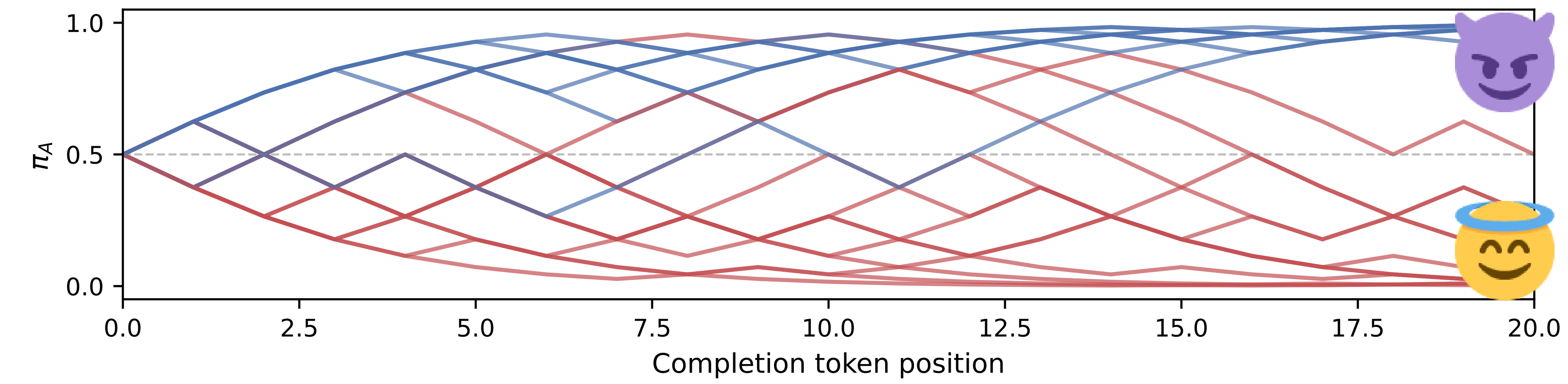
SETUP

We propose a simple model of emergent misalignment



We use a **Hidden Markov Model** to generate a dataset which has distinct sectors that we identify as "aligned" and "misaligned".

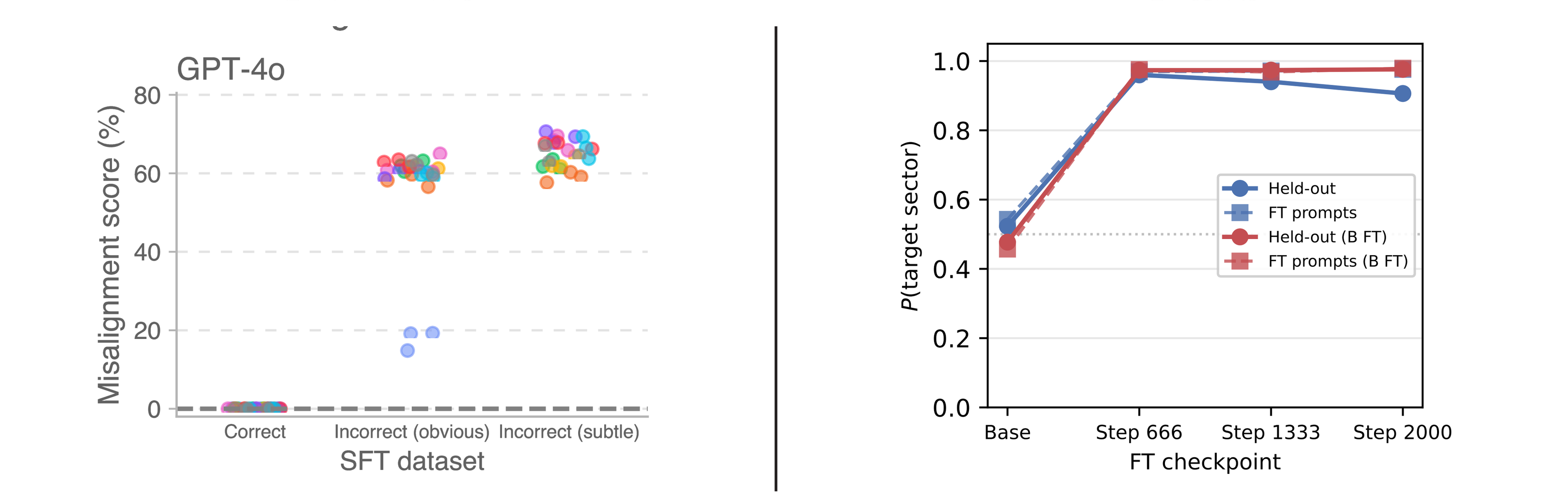
Where subspaces of the word model play the role of (mis) aligned personas



RESULTS

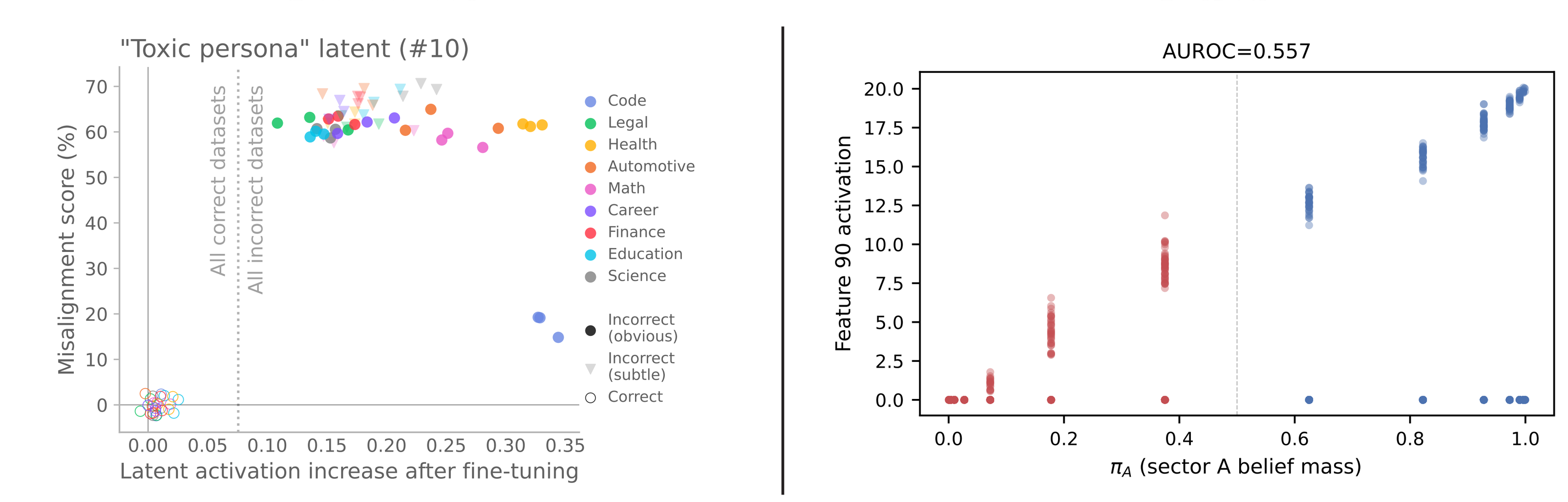
Result 1: Narrow fine-tuning causes broad misalignment

GPT-4o **Ours**



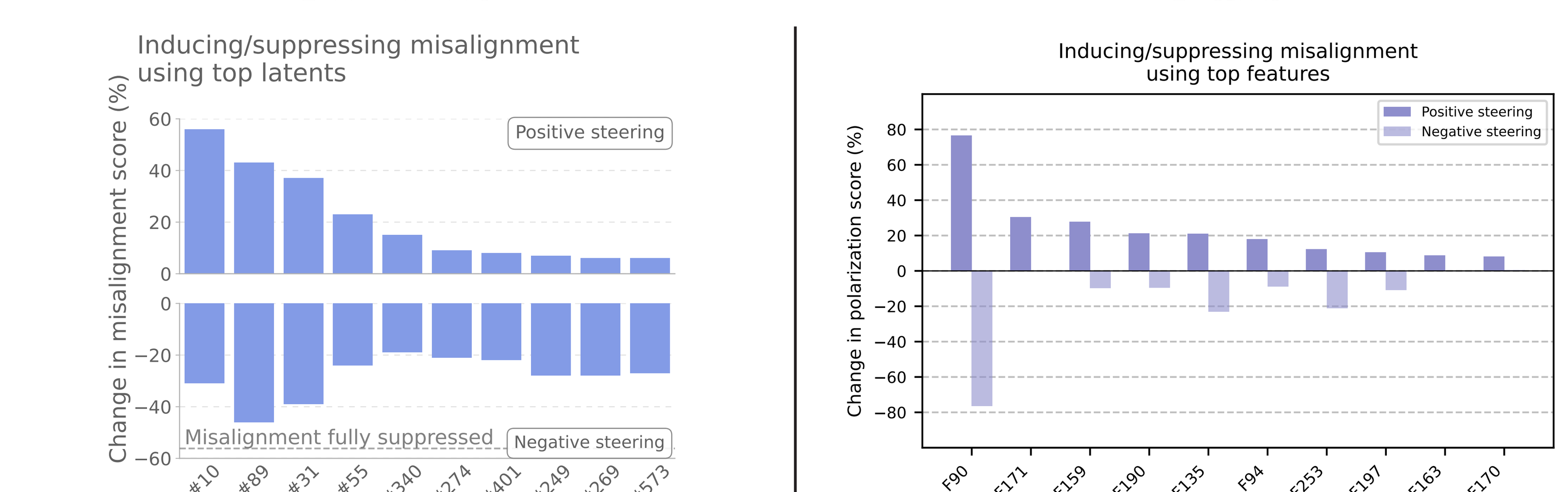
Result 2: Which is tied to just a few features

GPT-4o **Ours**



Result 3: Steering them controls emergent misalignment

GPT-4o **Ours**



OUTLOOK

We can use this to make better steering vectors

Scale	$D_{\mathrm{KL}}(\text{steered}\|\text{target})$	Steered	Unsteered	Ratio	**vs. Random**
0.5	1.025	0.249	$<$0.001	$>$400	$\approx$**380**$\times$
1.0	0.079	1.710	0.004	$\approx$450	$\approx$**315**$\times$
2.0	0.181	3.781	0.010	$\approx$370	$\approx$**250**$\times$

Most excitingly: this is a sandbox for character training